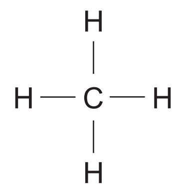
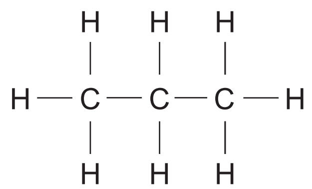
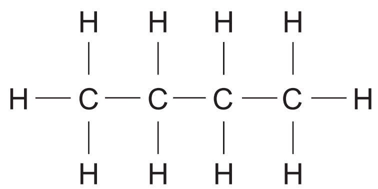
6. (a) The table shows the names, molecular formulae and structural formulae of some alkanes.

(i) Complete the table. [3]

Name	Molecular formula	Structural formula
…………………………………………	CH_4	
ethane	C_2H_6	
propane	…………………………………………	
butane	C_4H_{10}	

(ii) Give the **names** of the **two** elements present in all alkanes. [1]

………………………………………… and …………………………………………

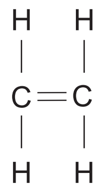
(iii) The general formula for the alkanes is C_nH_{2n+2} .

Circle the molecular formula for the alkane containing 6 carbon atoms. [1]

$\mathbf{C_6H_6}$ $\mathbf{CH_6}$ $\mathbf{C_6H_{12}}$ $\mathbf{C_6H_{14}}$

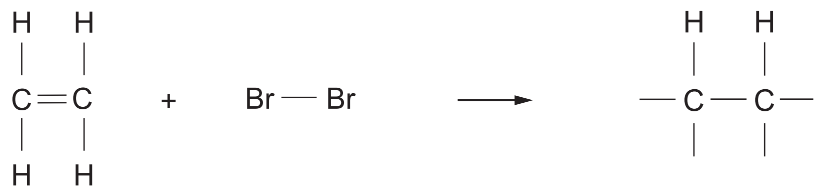
Examiner only

(b) Ethene belongs to a different family of hydrocarbons called the alkenes.

The structural formula of ethene is 

The presence of the double bond in alkenes can be identified using bromine water.

(i) Complete the structural formula for the product of the reaction between ethene and bromine. [1]



(ii) Complete the sentence below by underlining the correct words in the brackets. [2]

Ethene turns (**purple / orange / green / colourless**) bromine water

(**green / milky / colourless / blue**).

(c) Underline the name given to the type of reaction taking place when ethene reacts with bromine. [1]

addition **polymerisation** **oxidation** **neutralisation**

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